Lecture #5

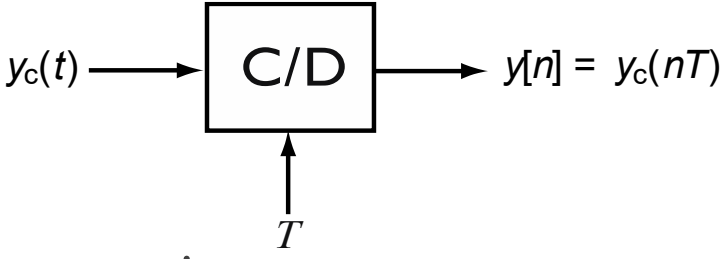
EE 471C / EE 381K-17 Wireless Communication Lab

Review from the previous lecture

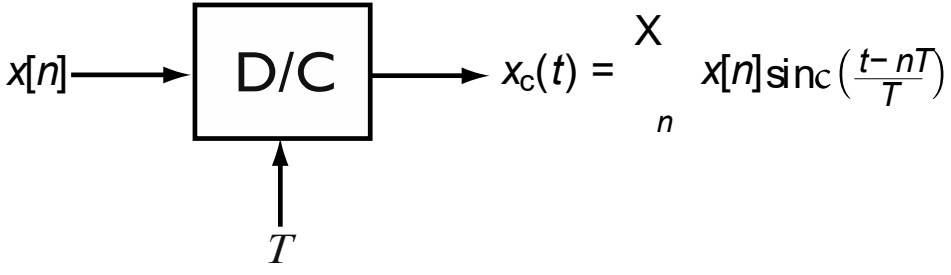
- Bandlimited signals are widely used in communication systems
- Bandlimited signals can be fully represented by their periodic samples if the period is small enough
- Digital signal processing (DSP) is important and relevant because of the use of bandlimited signals, DSP and communication are related

Graphical sampling notation

- C/D: continuous-to-discrete converter
  - Ideal analog-to-digital converter



- D/C: discrete-to-continuous converter
  - Ideal digital-to-analog converter



Preview of today's lecture

- Discrete-time Processing of Bandlimited Continuous-Time Signals
  - Determine equivalent combination of sampling, digital filtering, and reconstruction to process a bandlimited continuous-time signal with discrete-time signal processing.
- Defining the frequency response of a random signal
  - Define and calculate the power spectrum of a wide sense stationary stochastic process.
- Bandwidth of a signal
  - Define and calculate various real measures of bandwidth
- Complex envelope representation of wireless passband signals
  - Use the complex envelope to represent a passband signal at baseband.

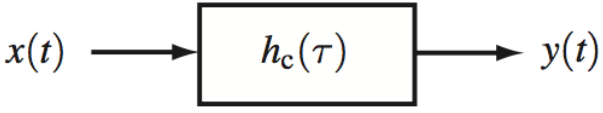
Discrete-time Processing of Bandlimited Continuous-Time Signals

Learning objective:

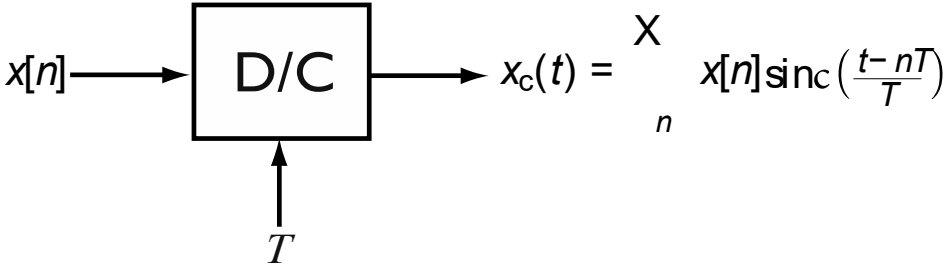
- Determine equivalent combination of sampling, digital filtering, and reconstruction to process a bandlimited continuous-time signal with discrete-time signal processing.

Filtering a bandlimited signal (1)

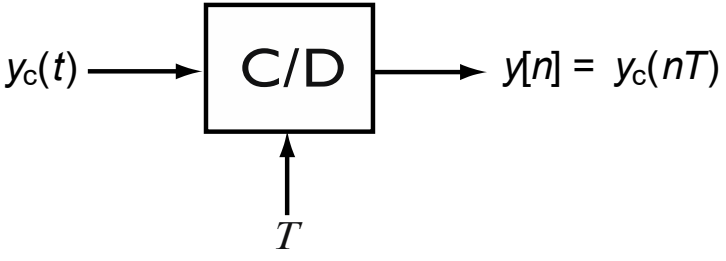
- Consider



- Because x(t) is bandlimited can generate it in discrete-time

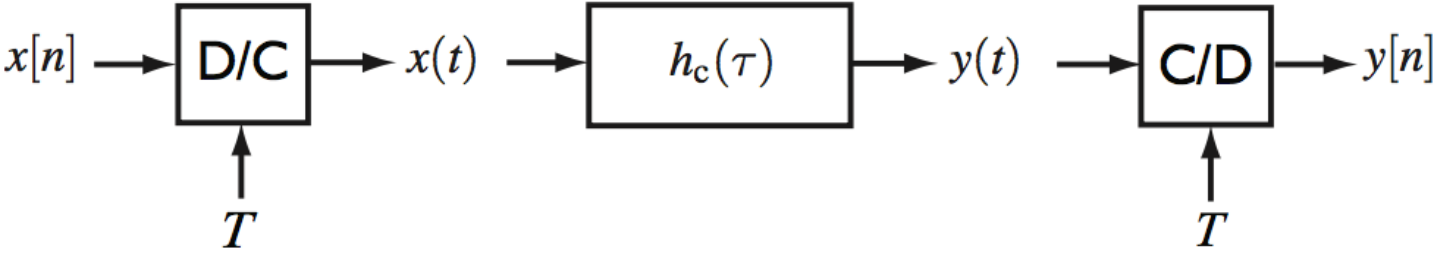


- Because y(t) is bandlimited can sample it to obtain discrete-time

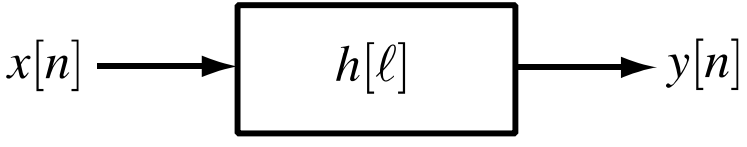


Filtering a bandlimited signal (2)

- Is it possible to replace

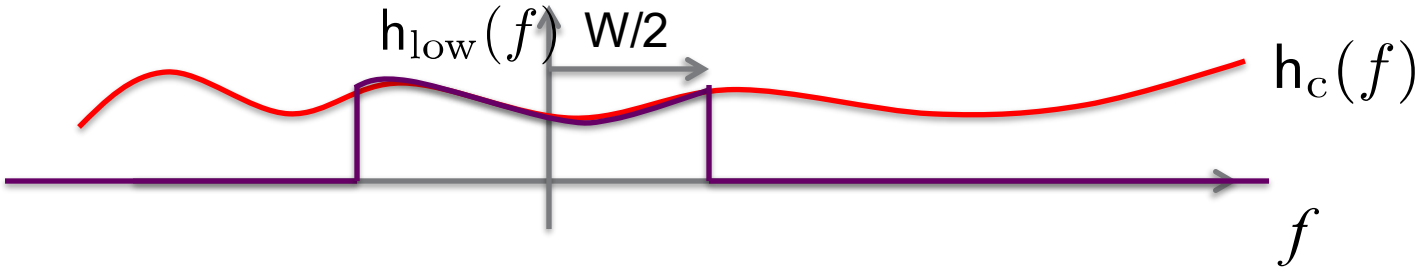


- With this for some suitable $h[n]$?



Yes!

Explaining the process (1)



- The portion of the channel in the bandwidth W/2

$$h_{low}(f) = \text{rect}(f/W)h_c(f)$$

- In the time domain

$$h_{low}(t) = W \int \text{sinc}(\tau W) h_c(t-\tau) \text{d}\tau$$

Explaining the process (2)

recall

$$X\left(e^{j2\pi f}\right) = \frac{1}{T}\sum_{n} X_c\left(\frac{f}{T} - \frac{n}{T}\right)$$

- The spectrum of $y_c(t)$

$$y_c(f) = h_{low}(f)x_c(f)$$

- The spectrum of $y[n]$

$$y\left(e^{j2\pi f}\right) = \frac{1}{T}y_c\left(\frac{f}{T}\right) \quad \text{for} \quad f \in [-1/2, 1/2)$$

$$= \frac{1}{T}h_{low}\left(\frac{f}{T}\right)x_c\left(\frac{f}{T}\right) \quad \text{for} \quad f \in [-1/2, 1/2)$$

Explaining the process (3)

- Substituting

$$\mathsf{y}\left(e^{\mathrm{j}2\pi f}\right) = \mathsf{h}_{\mathrm{low}}\left(\frac{f}{T}\right) \mathsf{x}\left(e^{\mathrm{j}2\pi f}\right) \quad \text{for} \;\; f \in [-1/2, 1/2)$$

$$= \frac{1}{T} T \mathsf{h}_{\mathrm{low}}\left(\frac{f}{T}\right) \mathsf{x}\left(e^{\mathrm{j}2\pi f}\right) \quad \text{for} \;\; f \in [-1/2, 1/2)$$

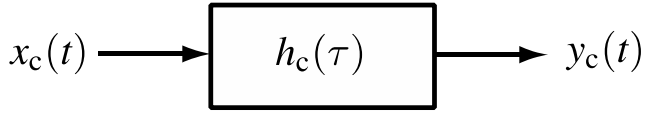
- Now suppose that

$$\mathsf{h}(f) = T\mathsf{h}_{\mathrm{low}}(f)$$

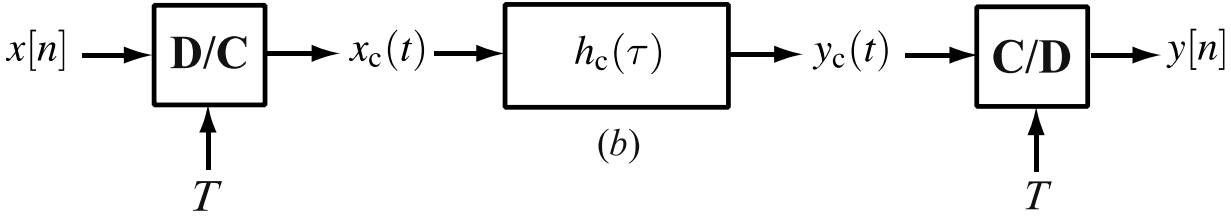
$$\mathsf{y}\left(e^{\mathrm{j}2\pi f}\right) = \frac{1}{T}\mathsf{h}\left(\frac{f}{T}\right) \mathsf{x}\left(e^{\mathrm{j}2\pi f}\right) \quad \text{for} \;\; f \in [-1/2, 1/2)$$

$$\mathsf{y}\left(e^{\mathrm{j}2\pi f}\right) = \mathsf{h}\left(e^{\mathrm{j}2\pi f}\right) \mathsf{x}\left(e^{\mathrm{j}2\pi f}\right) \quad \text{for} \;\; f \in [-1/2, 1/2)$$

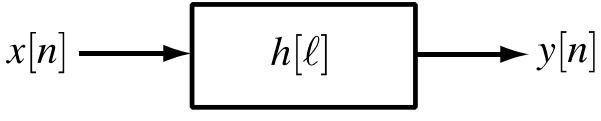
Discrete-time equivalente system



(a)



(b)



(c)

$$
\begin{aligned}
h[n] &= T h_{\text{low}}(nT) \\
&= TW \int \text{sinc}(\tau W) h_c(nT - \tau) \mathrm{d}\tau \\
&= \int \text{sinc}(\tau W) h_c(nT - \tau) \mathrm{d}\tau.
\end{aligned}
$$

TW=1 if sampling at exactly Nyquist

Example

- Compute the discrete-time equivalent system if

$$h(t) = \delta(t - \tau_0) + \alpha\delta(t - \tau_1)$$

Discrete-time Processing of Bandlimited Continuous-Time Signals

Learning objective:

- Determine equivalent combination of sampling, digital filtering, and reconstruction to process a bandlimited continuous-time signal with discrete-time signal processing.

Defining the frequency response of a random signal

Learning objective:

- Define and calculate the power spectrum of a wide sense stationary stochastic process.

Random processes in the frequency domain

- Random signals do not have a conventional definition of a spectrum
  - If x(t) is a random process,
    - What does the Fourier transform of x(t) mean?
    - Is it well defined?
- The power spectrum is the Fourier transform of the autocovariance function of a WSS random process
  - Use power spectrum to define **bandwidth** of a random signal
  - Use power spectrum to compute **power** of a random signal

The power spectrum for random processes

- For continuous-time signals

$$P_x(f) = \int_{-\infty}^{\infty} C_{xx}(t) e^{-\mathrm{j}2\pi f t} \mathrm{d}t$$

- For discrete-time signals

$$P_x(e^{\mathrm{j}2\pi f}) = \sum_{n=-\infty}^{\infty} C_{xx}[n] e^{-\mathrm{j}2\pi f n}$$

- Power spectrum is real and non-negative due to conjugate symmetry

Exercise

- Find the power spectrum for a zero-mean WSS random process x(t) with an exponential correlation function with

$$R_{xx}(t) = e^{-2\beta|t|} \qquad t \in (-\infty, \infty)$$

Step 1: Use the Wiener–Khinchin Theorem

The **power spectral density** $S_{xx}(f)$ is the **Fourier transform** of the autocorrelation function:

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(t) e^{-j2\pi ft} dt$$

Substitute $R_{xx}(t) = e^{-2\beta|t|}$:

$$S_{xx}(f) = \int_{-\infty}^{\infty} e^{-2\beta|t|} e^{-j2\pi ft} dt$$

Step 2: Split the Integral

Because of the absolute value:

$$S_{xx}(f) = \int_{-\infty}^{0} e^{2\beta t} e^{-j2\pi f t} dt + \int_{0}^{\infty} e^{-2\beta t} e^{-j2\pi f t} dt$$

Evaluate each part:

- For $t < 0$:

$$\int_{-\infty}^{0} e^{(2\beta - j2\pi f)t} dt = \frac{1}{2\beta - j2\pi f}$$

- For $t > 0$:

$$\int_{0}^{\infty} e^{-(2\beta + j2\pi f)t} dt = \frac{1}{2\beta + j2\pi f}$$

Step 3: Add the Two Parts

So the PSD is:

$$S_{xx}(f) = \frac{1}{2\beta - j2\pi f} + \frac{1}{2\beta + j2\pi f}$$

Combine:

$$S_{xx}(f) = \frac{(2\beta + j2\pi f) + (2\beta - j2\pi f)}{(2\beta)^2 + (2\pi f)^2} = \frac{4\beta}{4\beta^2 + 4\pi^2 f^2}$$

Simplify:

$$S_{xx}(f) = \frac{\beta}{\beta^2 + \pi^2 f^2}$$

Defining the frequency response of a random signal

Learning objective:

- Define and calculate the power spectrum of a wide sense stationary stochastic process.

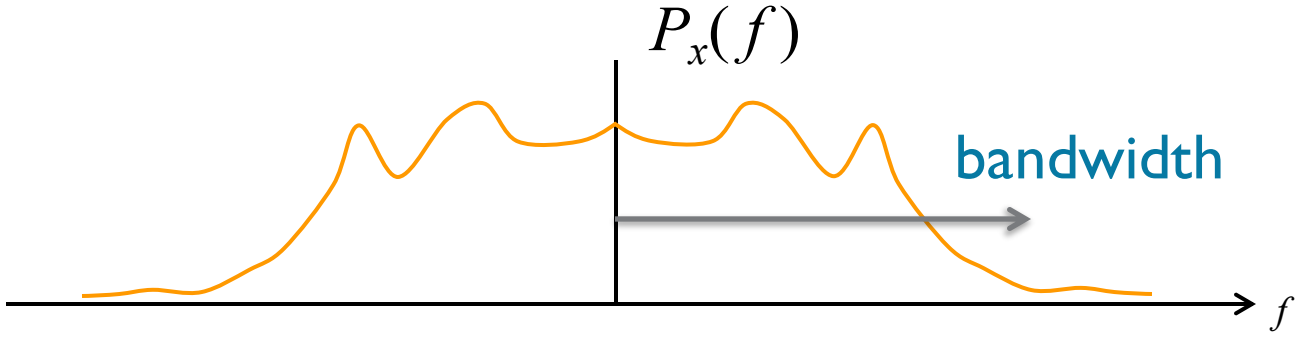
Break

Bandwidth of a signal

Learning objective:

- Define and calculate various real measures of bandwidth

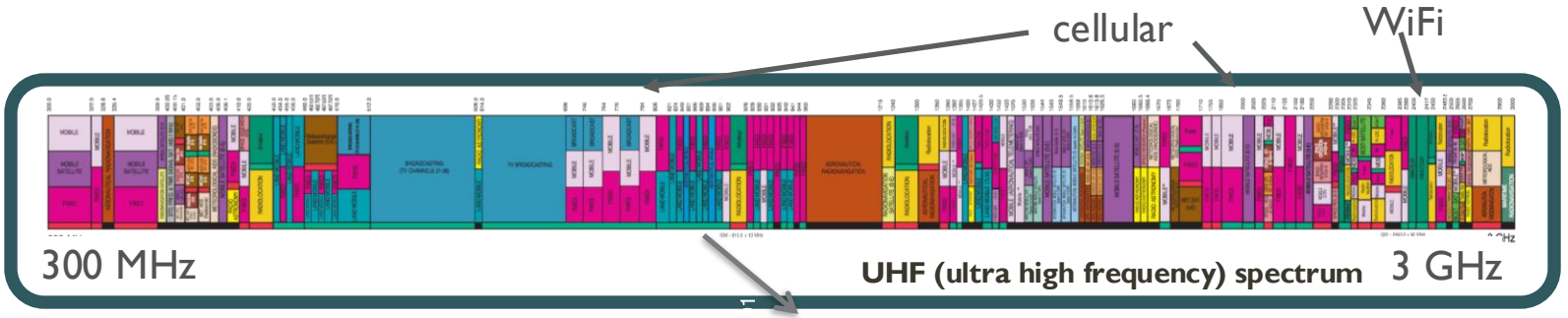
Bandwidth of a signal



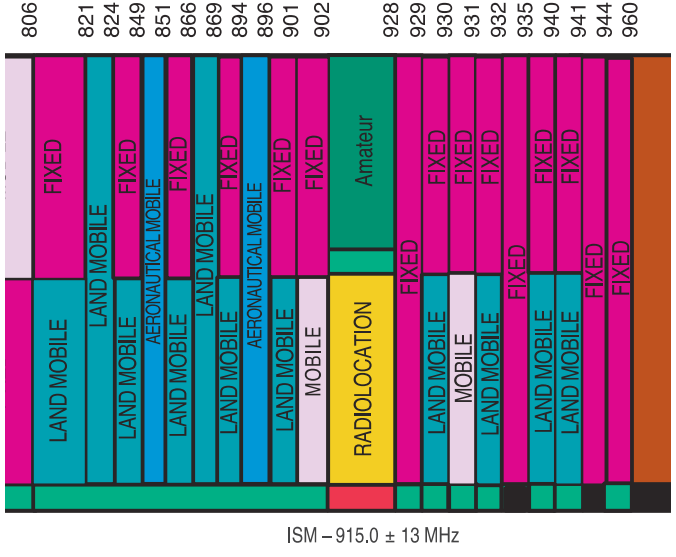
This is a **baseband** signal

- Bandwidth is determined from its Fourier transform or power spectrum
- Bandwidth is some measure of the extent of frequency content

Spectrum allocation in wireless

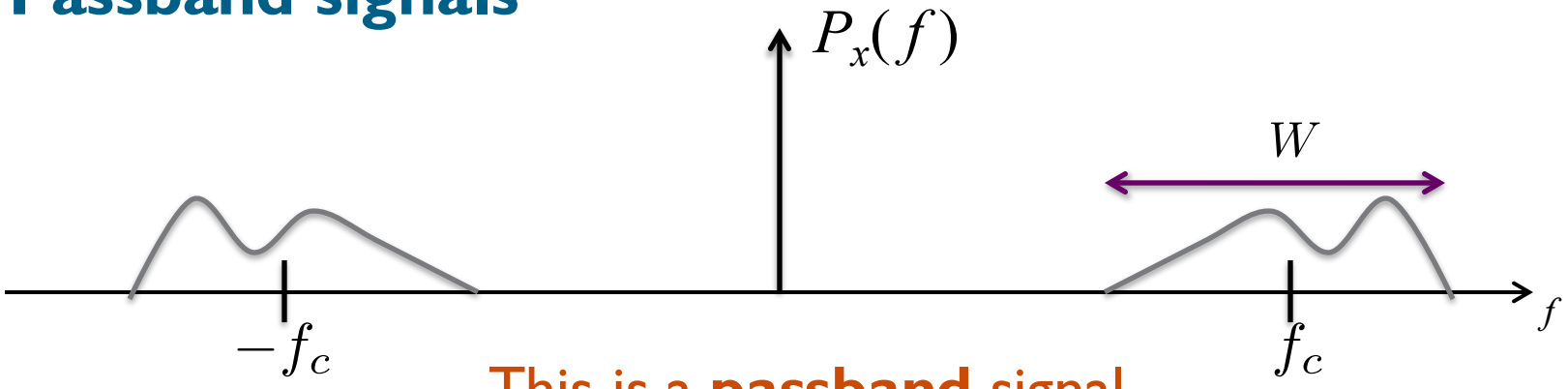


Wireless services occupy small slices of larger frequency spectrum



FYI: Cellular is land mobile

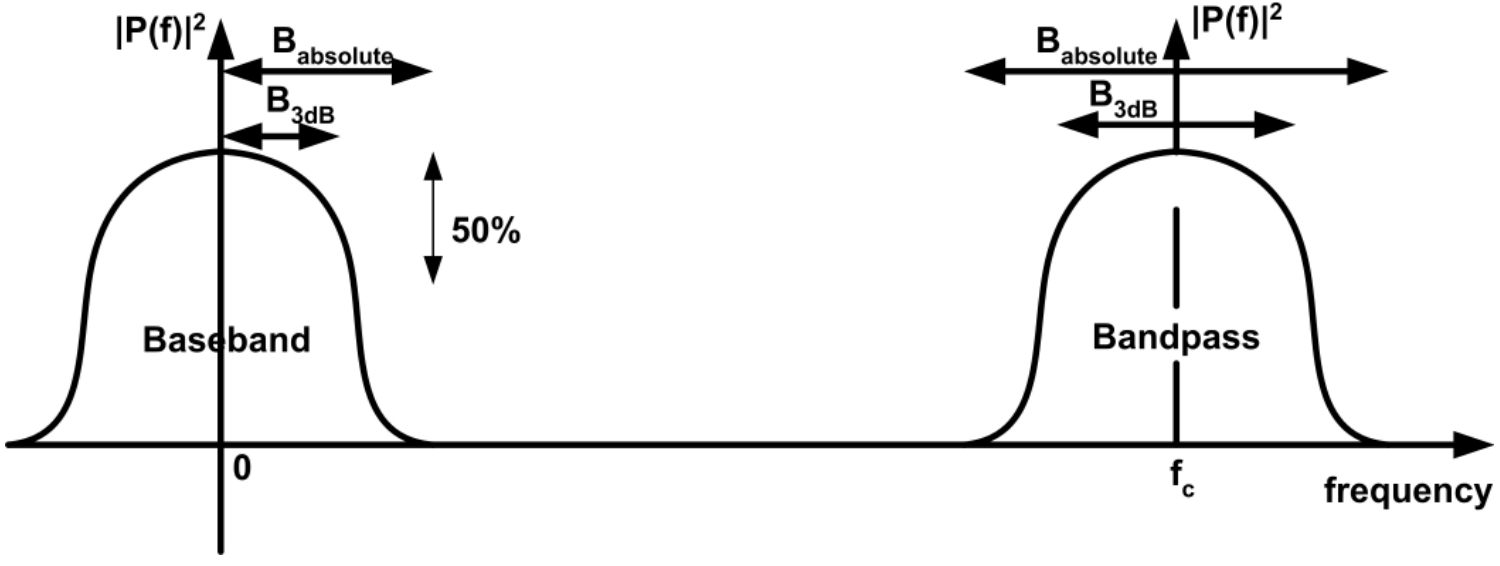
Passband signals



This is a **passband** signal

- **Passband signals** have energy that is nonzero for a frequency band concentrated about +/-f_c , where f_c is the carrier frequency and $f_c \gg 0$.
- Bandwidth is determined by signal extent around the carrier
  - All wireless signals are passband
  - There are some variations of this definition that place the carrier differently

Measuring the bandwidth



- Several different criteria for measuring bandwidth
- Definition depends on whether baseband or passband

Some common measures

Half-power bandwidth The half-power bandwidth, or 3dB bandwidth, is defined as the value of the frequency over which the power spectrum is at least 50% of its maximum value, i.e.

$$B = f_{3\text{dB}}^{(h)} - f_{3\text{dB}}^{(\ell)}, \tag{160}$$

where $f_{3\text{dB}}^{(h)}$ and $f_{3\text{dB}}^{(\ell)}$ are as shown in Fig. 2.

XdB bandwidth The XdB bandwidth is defined as the difference between the largest frequency that suffers XdB of attenuation and the smallest frequency. Similar to the half-power bandwidth, this can be expressed as,

$$B = f_{X\text{dB}}^{(h)} - f_{X\text{dB}}^{(\ell)}, \tag{161}$$

where $f_{X\text{dB}}^{(h)}$ and $f_{X\text{dB}}^{(\ell)}$ are defined in the same manner as $f_{3\text{dB}}^{(h)}$ and $f_{3\text{dB}}^{(\ell)}$.

Noise equivalent bandwidth The noise equivalent bandwidth is defined as

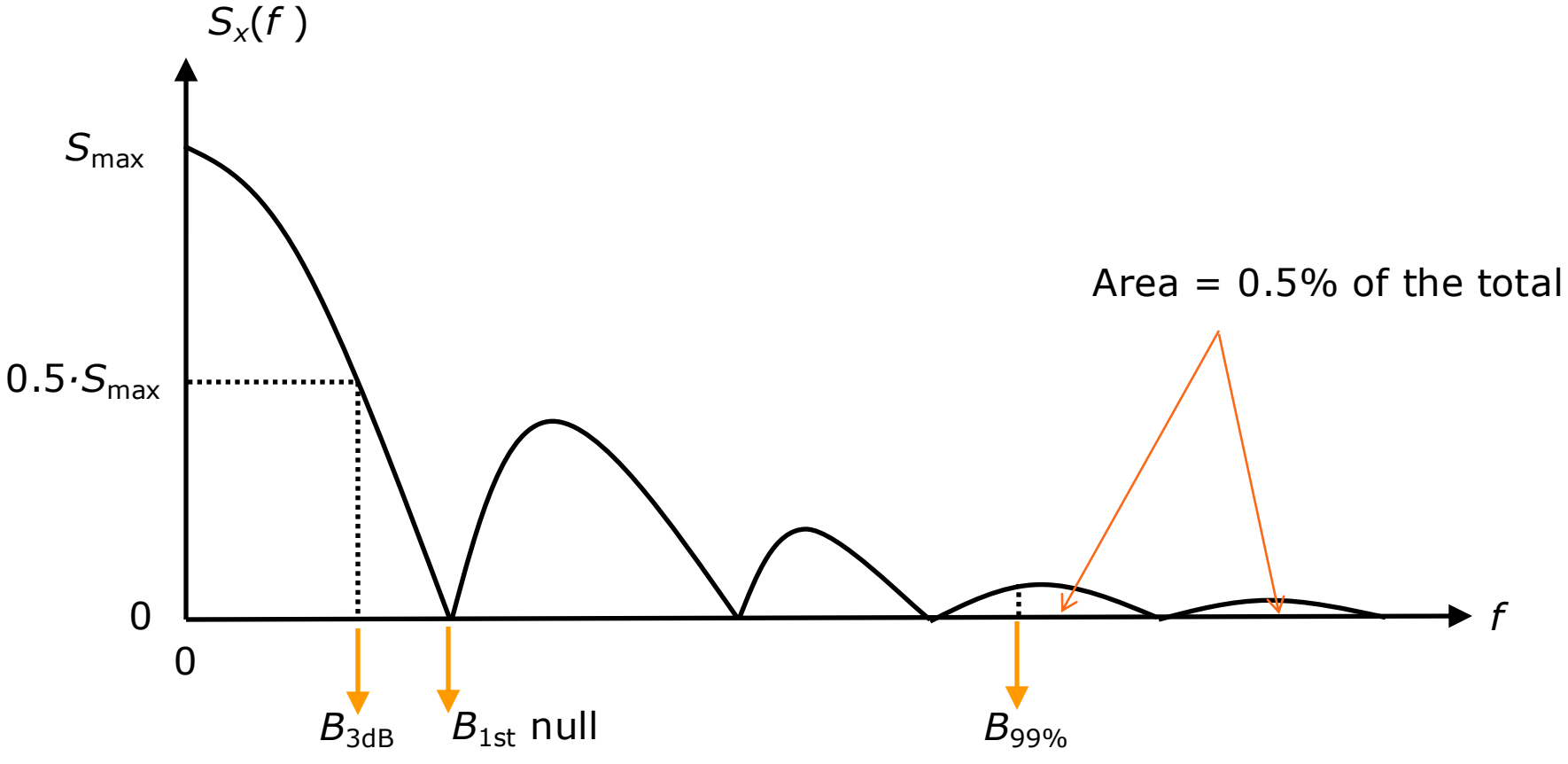
$$B = \frac{1}{P_x(f_\text{c})} \int_f P_x(f)\text{d}f. \tag{162}$$

Fractional power containment bandwidth One of the most useful notions of bandwidth, often employed by the FCC, is the fractional power containment bandwidth. For a passband signal, this bandwidth is defined as the B that satisfies

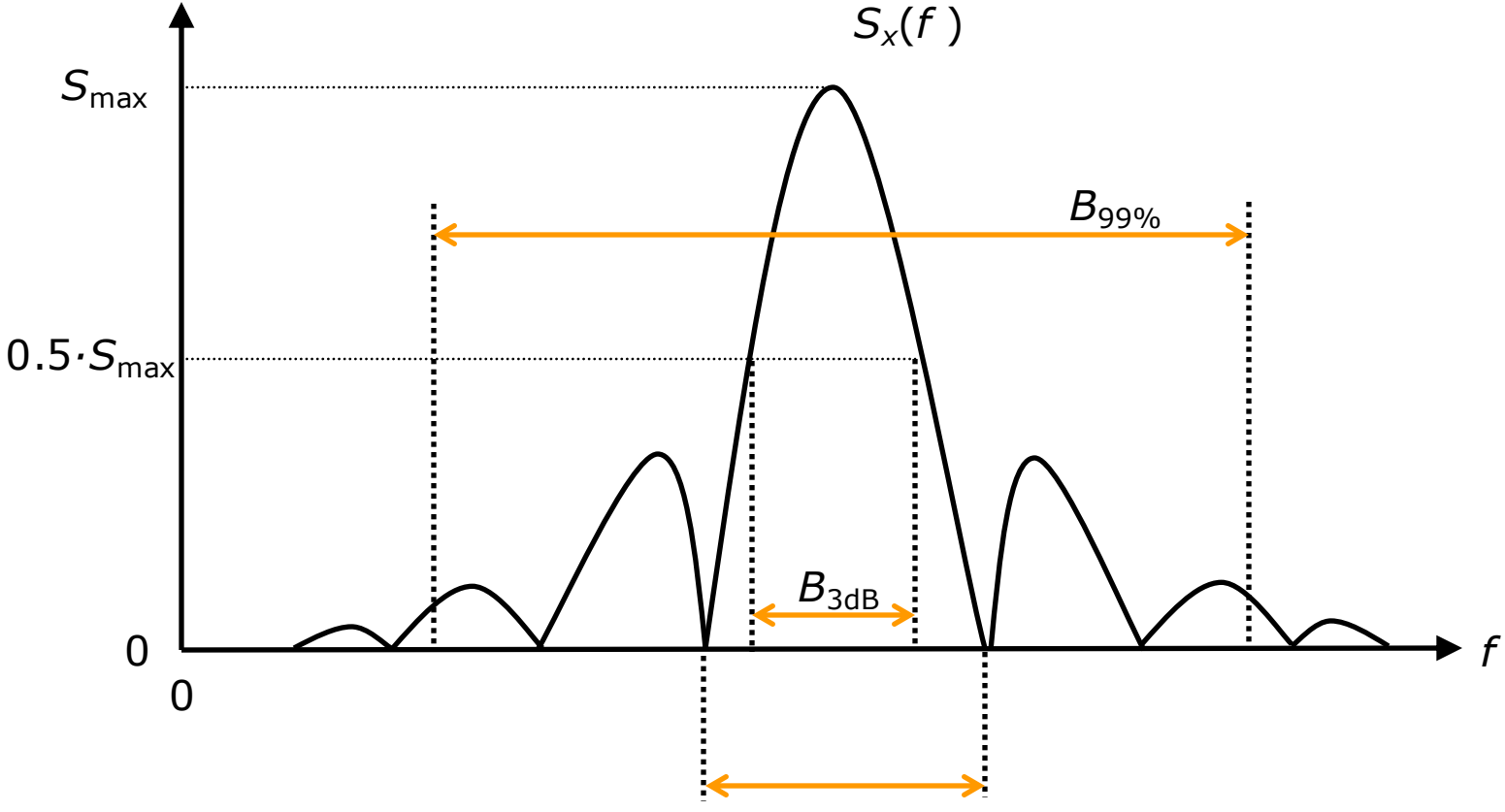
$$\int_{f_\text{c}-B/2}^{f_\text{c}+B/2} P_x(f)\text{d}f = \alpha \int_{-\infty}^{\infty} P_x(f)\text{d}f \tag{163}$$

where α is the fraction of containment. For example, if $\alpha = 0.99$ then the bandwidth B would be defined such that the signal has $\alpha\%$ of its bandwidth.

Baseband illustration



Passband illustration



Bandwidth of a signal

Learning objective:

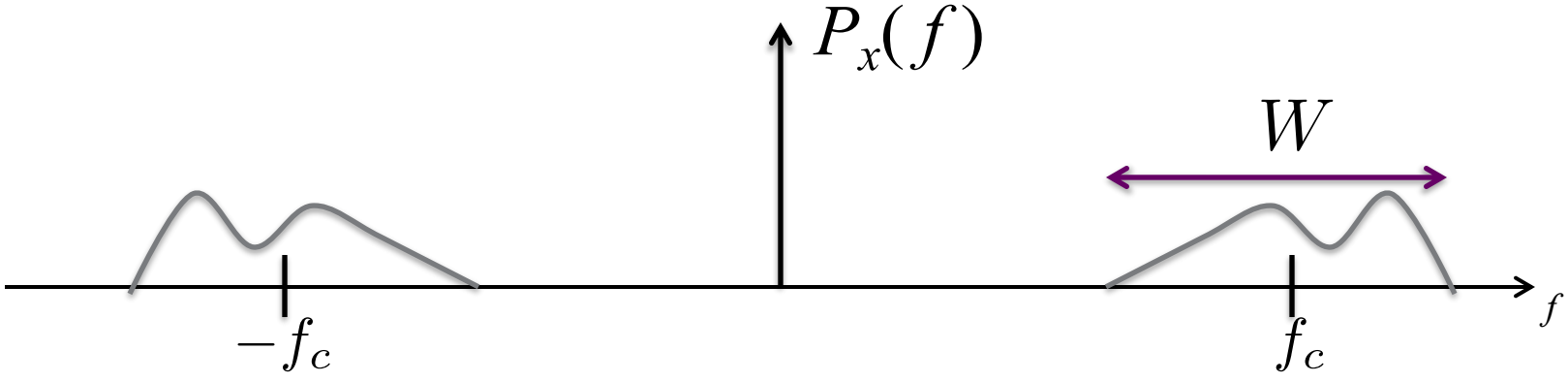
- Define and calculate various real measures of bandwidth

Complex envelope representation of wireless passband signals

Learning objective:

- Use the complex envelope to represent a passband signal at baseband

Wireless passband signals

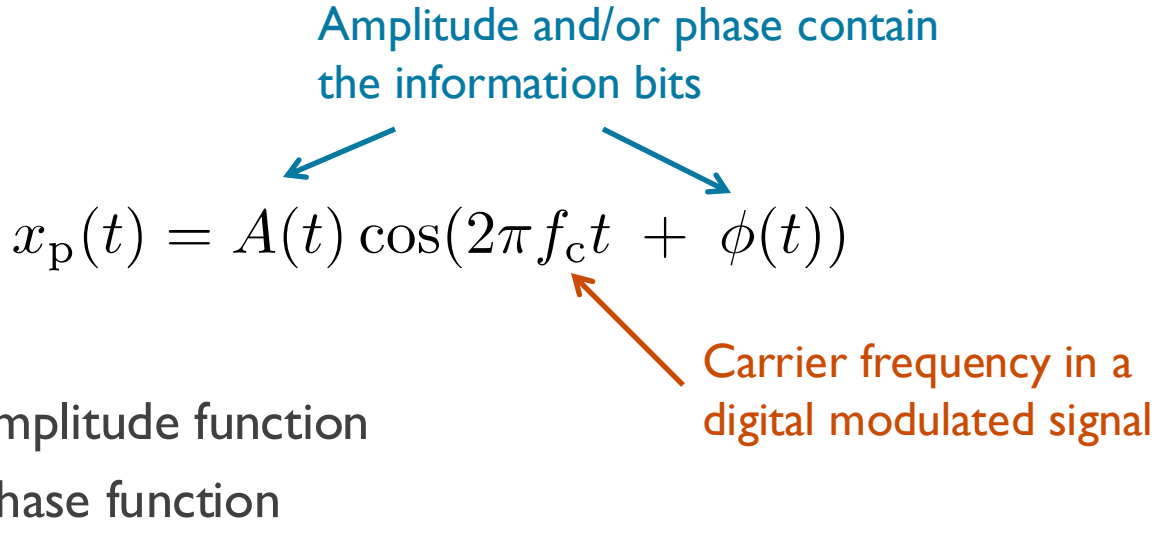


- **Passband signals** have energy that is nonzero for a frequency band concentrated about the carrier f_c
- Most passband wireless signals are also narrowband

$$W \ll f_c$$

Passband signal and simplifications

- General passband signal can be written



Inphase and quadrature signals

- Apply the trig identity $\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$

$$x_{\mathrm{p}}(t) = \underbrace{A(t)\cos(\phi(t))}_{x_{\mathrm{i}}(t)}\cos(2\pi f_{\mathrm{c}} t) - \underbrace{A(t)\sin(\phi(t))}_{x_{\mathrm{q}}(t)}\sin(2\pi f_{\mathrm{c}} t)$$

inphase quadrature

- Complex envelope $x(t) = x_{\mathrm{i}}(t) + \mathrm{j}x_{\mathrm{q}}(t)$

Complex envelope to passband signal

- Multiplying

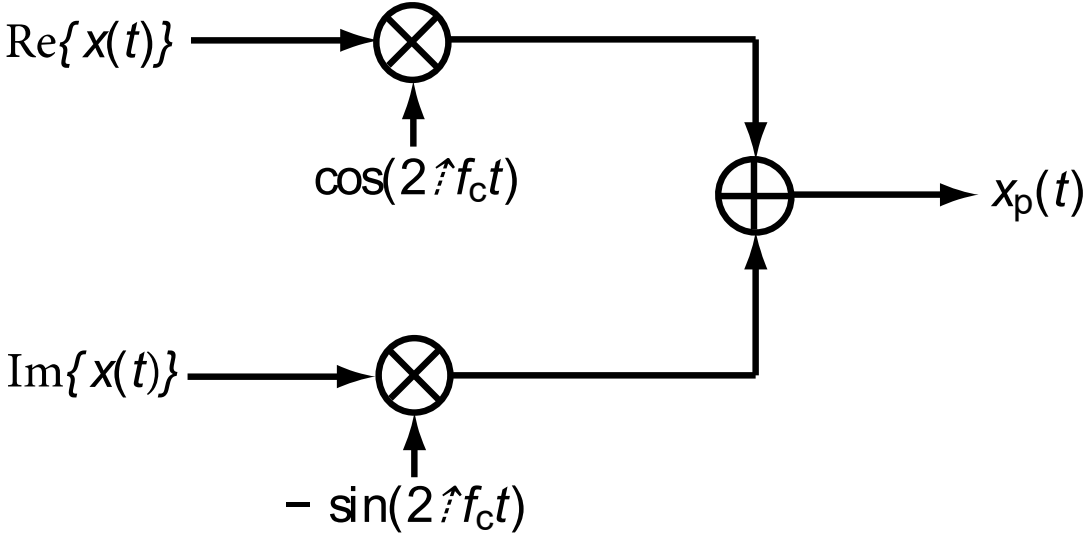
$$x(t)e^{\mathrm{j}2\pi f_c t} = x_{\mathrm{i}}(t)\cos(2\pi f_c t) - x_{\mathrm{q}}(t)\sin(2\pi f_c t) + \mathrm{j}(x_{\mathrm{i}}(t)\sin(2\pi f_c t) + x_{\mathrm{q}}(t)\cos(2\pi f_c t))$$

- Therefore

$$x_p(t) = \mathrm{Re}\left[x(t)e^{\mathrm{j}2\pi f_c t}\right]$$

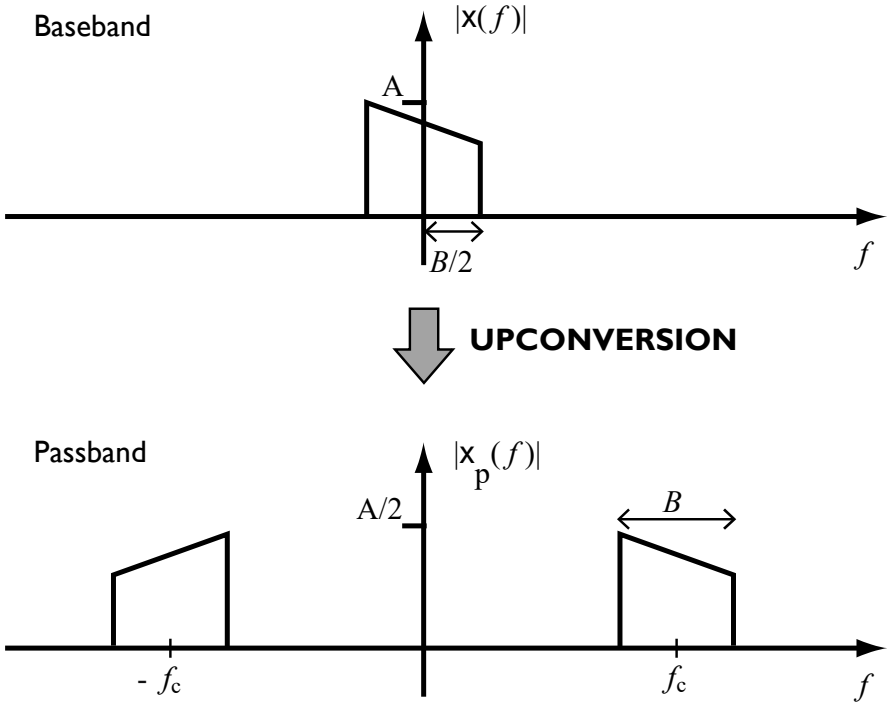
UPCONVERSION

Upconversion



- Process of going from complex envelope to passband

Upconversion visualized



- In the time domain

$$x_p(t) = \mathrm{Re}\left[x(t)e^{j2\pi f_c t}\right]$$

or

$$x_p(t) = \frac{1}{2}x(t)e^{j2\pi f_c t} + \frac{1}{2}x^*(t)e^{-j2\pi f_c t}$$

- In the frequency domain

$$x_p(f) = \frac{1}{2}\left(x(f - f_c) + x^*(-f - f_c)\right)$$

Passband signal to baseband

- Consider a passband signal $y_p(t)$
- How to obtain the complex envelope from a passband signal?

$$y_p(t) = \mathrm{Re}\left[y(t)e^{j2\pi f_c t}\right]$$



DOWNCONVERSION

Using trig identities

- Consider the identities

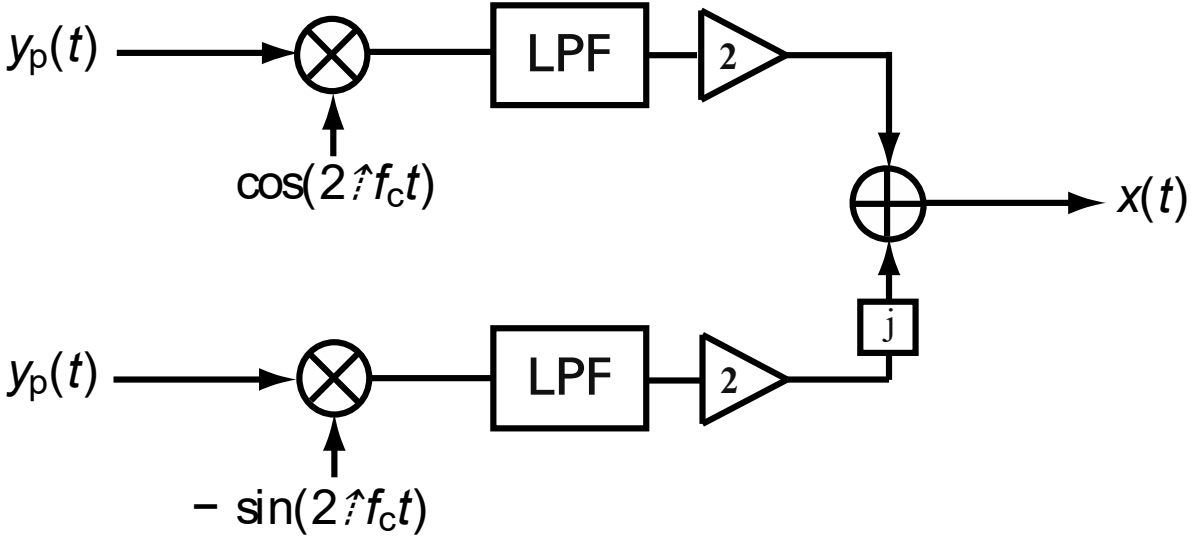
$$\sin u \sin v = \frac{1}{2}\left[\cos(u-v) - \cos(u+v)\right]$$
$$\cos u \cos v = \frac{1}{2}\left[\cos(u-v) + \cos(u+v)\right]$$
$$\sin u \cos v = \frac{1}{2}\left[\sin(u-v) + \sin(u+v)\right]$$

- Applying the identities

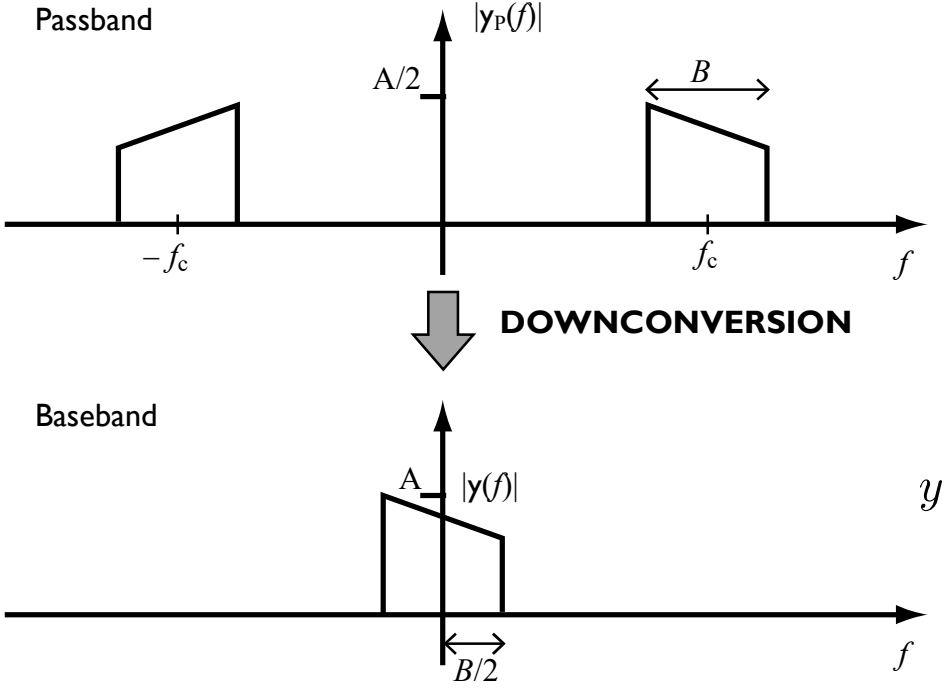
$$y_{\mathrm{p}}(t)\cos(2\pi f_{\mathrm{c}} t) = \frac{1}{2}y_{\mathrm{i}}(t) + \boxed{\frac{1}{2}y_{\mathrm{i}}(t)\cos(4\pi f_{\mathrm{c}} t) - \frac{1}{2}y_{\mathrm{q}}(t)\sin(4\pi f_{\mathrm{c}} t)}$$
$$y_{\mathrm{p}}(t)\sin(2\pi f_{\mathrm{c}} t) = -\frac{1}{2}y_{\mathrm{q}}(t) \boxed{+ \frac{1}{2}y_{\mathrm{q}}(t)\cos(4\pi f_{\mathrm{c}} t) + \frac{1}{2}y_{\mathrm{i}}(t)\sin(4\pi f_{\mathrm{c}} t)}$$

Filter out

Downconversion



Downconversion visualized



- In the frequency domain

$$\mathrm{y}(f) = 2\,\mathrm{rect}\left(\frac{f}{W}\right)\mathrm{y_p}(f - f_\mathrm{c})$$

- In the time domain

$$y(t) = 2W\int \mathrm{sinc}((t-\tau)W)y_\mathrm{p}(t)e^{-\mathrm{j}2\pi f_\mathrm{c} t}\mathrm{d}\tau.$$

Complex envelope representation of wireless passband signals

Learning objective:

- Use the complex envelope to represent a passband signal at baseband